

AN2DL - Second Challenge Report

DarkHorse

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1 Introduction

This project addresses a multiclass *histopathological image classification* task aimed at predicting breast cancer molecular subtypes from microscopic tissue images. Each sample is associated with a binary segmentation mask highlighting regions likely to contain diseased tissue. A key difficulty lies in the **limited size and quality of the dataset**. To address these constraints, our approach combines robust preprocessing with mask-guided region selection and convolutional neural networks to extract discriminative features from tissue-relevant areas.

2 Problem Analysis

Dataset Characteristics

The dataset consists of 1,168 histopathological images of varying resolutions, each paired with a binary mask. The original training set contained 691 labeled samples across four molecular subtypes, with 477 unlabeled samples provided for testing.

Preliminary inspection revealed substantial data quality issues. In particular, 110 training samples were discarded due to duplication or severe artifacts that compromised both image quality and mask reliability. After cleaning, the final training set comprised 581 labeled images with the following distribution:

Luminal B: 35.1%

Luminal A: 27.2%

HER2-positive (HER2(+)): 25.8%

Triple Negative (TNBC): 11.9%

This distribution shows moderate class imbalance, with Triple Negative significantly underrepresented.

Challenges and Assumptions

The dataset poses several challenges typical of medical imaging tasks under limited supervision. With only 581 usable training samples, the risk of overfitting is substantial, necessitating strong regularization, extensive data augmentation, and robust validation strategies. Additionally, non-uniform staining and scanning procedures introduce significant variability in color distribution, resolution, and visual appearance, complicating the learning of consistent discriminative features.

Finally, quantitative analysis showed that the segmentation masks are extremely sparse (median coverage $\approx 1\%$) and provide limited discriminative signal, motivating their use solely for ROI extraction rather than direct supervision.

3 Methods

The proposed methodology addresses the challenges identified in Section 2, inspired by recent work in computational pathology [5, 2].

3.1 Data Preprocessing and Preparation

To reduce non-biological variability and ensure diagnostically relevant input, a dedicated preprocessing pipeline was applied. Images were standardized using **Reinhard color normalization**, aligning LAB channel statistics to a reference slide to mitigate H&E staining and scanner variability, thereby encouraging the model to focus on morphological rather than chromatic cues. Segmentation masks were used indirectly for **ROI extraction**, cropping images around tissue regions and resizing them to a fixed resolution to remove background and uninformative areas.

3.2 Training Strategies & Regularization

Given the critically small dataset size, strong regularization and careful training design were essential to ensure stable learning and prevent overfitting.

The cleaned dataset was partitioned using **stratified sampling**, and stratified cross-validation (CV) was employed during development to obtain more reliable performance estimates and reduce variance.

Data augmentation was applied through random geometric transformations (horizontal/vertical flips and rotations) to increase sample diversity while preserving tissue morphology. In addition, **MixUp regularization** was employed to generate virtual training examples via convex combinations of image-label pairs, encouraging smoother decision boundaries and improving generalization in the low-sample regime.

Class imbalance was addressed through **weighted random sampling**, oversampling underrepresented classes—particularly Triple Negative cases—to ensure balanced exposure across minibatches. Training was further stabilized using **early stopping** based on validation performance.

3.3 Model Architecture

Instead of a flat four-class classifier, we adopted a **hierarchical cascade of three binary classifiers** [2].

The first classifier C_1 performs a coarse triage between *Luminal* (Luminal A, Luminal B) and *Aggressive* (HER2(+), Triple Negative) subtypes. This grouping reflects clinically meaningful differences in treatment pathways, but is also motivated

by significant morphological and textural differences in the histopathological images (e.g., cell arrangement, nuclear size, and H&E staining structure). Since all downstream decisions depend on this initial split, its reliability is critical for the overall system.

Conditioned on the triage outcome, C_2 discriminates between Luminal A and Luminal B, while C_3 separates HER2(+) from Triple Negative cases. This decomposition simplifies each learning task and improves robustness under limited supervision.

3.4 Ensembling & TTA

Final predictions were obtained using **model ensembling combined with test-time augmentation (TTA)**. For each classifier (C_1, C_2, C_3), the five best models from CV were ensembled, and predictions were averaged across multiple augmented views (original, horizontal flip, vertical flip, rotation). This strategy reduces variance and improves robustness under stochastic training effects.

4 Experiments

Due to the limited size of the labeled dataset, extensive CV was employed during development, while final model selection was guided by leaderboard (LB) performance, which better reflects generalization.

Baselines and Early Experiments

Initial experiments employed **single-stage multi-class classifiers** with ImageNet-pretrained backbones (ConvNeXt-Tiny, EfficientNet-B0, ResNet50) trained on globally resized images. Although these models achieved moderate validation performance, they consistently underperformed on the leaderboard (Macro F1 ≈ 0.27), revealing poor generalization. This exposed a fundamental limitation of global resizing, which fails to capture localized morphological patterns.

To address this limitation, **patch-based training** was introduced, extracting fixed-size tiles and aggregating predictions using mean or top-K pooling. While this improved validation scores, LB performance remained unstable, revealing strong sensitivity to background and low-information tiles.

MIL-Based Approaches

Multiple Instance Learning (MIL) frameworks were explored to better capture intra-image heterogeneity. Among MIL approaches, **Attention-Based MIL (AB-MIL)** [1] with ConvNeXt-Tiny significantly improved cross-validation metrics and yielded one of the first strongest LB results. However, several extensions—including dual-stream RGB+mask architectures, FiLM-based conditioning (*Feature-wise Linear Modulation* [4]), and explicit mask-feature fusion—failed to translate into LB gains, suggesting that mask information was too sparse and noisy to support direct semantic learning.

Model Averaging Across Random Seeds

To further reduce variance induced by stochastic training factors, we employed **average seeding**. The training procedure was repeated using three distinct random seeds; for each seed, a 5-fold stratified CV was performed, resulting in a total of 15 trained models. The final prediction was obtained by averaging the predicted probabilities across all models, in combination with test-time augmentation.

Models trained under slightly different initial conditions tend to make complementary errors; therefore, probability averaging effectively smooths decision boundaries and consistently improved macro F1-score stability on the LB.

Hierarchical Classification

Finally, motivated by clinical taxonomy and prior work, a **hierarchical classification pipeline** composed of three binary classifiers was implemented, as detailed in Section 3.3. This decomposition simplified each decision step and reduced systematic class confusion.

5 Results

Table 1 summarizes the key LB results. Several technically sophisticated approaches—such as dual-stream models, multi-stage training and mask-as-channel architectures—failed to improve performance, underscoring that increased architectural complexity alone is ineffective when supervision is weak. In contrast, strategies emphasizing problem

decomposition, robust aggregation, and inference-time constraints consistently yielded superior generalization.

Table 1: Representative Leaderboard F1 scores.

Method	LB F1
Single-stage CNN (global resize)	~0.27
Patch-based CNN + Mean Pooling	~0.31
AB-MIL (ConvNeXt-Tiny)	0.3668
Dual-Stream RGB+Mask	0.3440
Mask-Guided EffNet-B0 (3-Seed Avg + TTA)	0.3853
Hierarchical Binary Cascade (final)	0.3978

6 Discussion

The experimental results confirm that breast cancer molecular subtyping from histopathological images under severe data constraints is primarily limited by dataset size and intrinsic variability rather than by model capacity. Increasing architectural complexity consistently led to higher variance and overfitting, while performance improvements were mainly obtained through data-centric choices. In particular, segmentation masks proved ineffective for direct supervision, hierarchical decomposition and model averaging improved robustness by reducing background sensitivity and systematic class confusion.

7 Conclusions

We presented a comprehensive pipeline for breast cancer molecular subtyping under severe data limitations. The main contributions include robust preprocessing, indirect mask utilization, hierarchical binary decomposition, and extensive ensembling with test-time augmentation.

Further gains are unlikely to arise from increased architectural complexity alone. Instead, future work should focus on **effective data expansion**, such as aggressive multi-scale tiling and tissue-driven patch extraction, and on leveraging self-supervised or pathology-pretrained encoders [5]. These directions offer the most promising path forward in low-data medical imaging settings.

A Appendix

This appendix reports supplementary experiments, negative results, and methodological reflections that could not be included in the main paper due to space constraints. These analyses guided our final design choices and provide further insight into the limitations imposed by the dataset.

A.1 Analysis of Segmentation Mask Utility

Given the availability of binary segmentation masks, we conducted a dedicated sanity analysis to assess their informativeness. We examined mask coverage statistics, geometric features (area, density, roughness), and low-dimensional embeddings of mask-derived features across classes.

Mask coverage and distribution. Overall mask coverage is extremely low, with a median of approximately 1% and a strongly right-skewed distribution (Figure 1). When stratified by molecular subtype, coverage distributions show substantial overlap and no statistically significant differences (Kruskal–Wallis $p \approx 0.84$), indicating that mask size alone does not encode subtype-specific information.

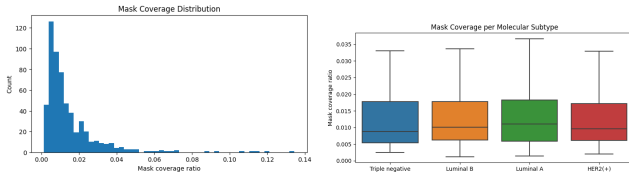


Figure 1: Left: global mask coverage distribution. Right: mask coverage stratified by molecular subtype.

Mask geometry and shape features. Simple geometric descriptors (area, density, roughness, compactness) were evaluated to assess whether mask shape correlates with subtype. Compactness distributions overlap strongly across classes, and a t-SNE embedding of low-order geometry features forms a single mixed manifold with no class separation (Figure 2), suggesting weak semantic signal.

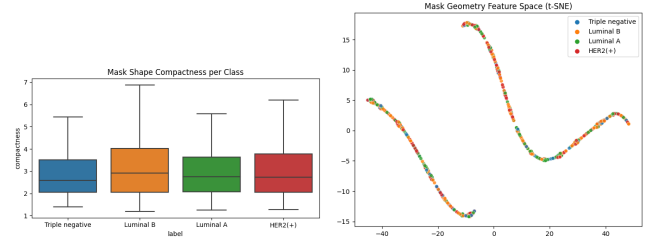


Figure 2: Left: mask compactness per subtype. Right: t-SNE embedding of basic mask geometry features.

Conclusion. These analyses indicate that while segmentation masks are spatially well-aligned and technically valid, they carry limited subtype-specific information. Their primary value lies in indirect usage—such as guiding region-of-interest extraction and patch sampling—rather than direct semantic supervision or standalone prediction.

A.2 Comparison with Alternative Pipelines from Literature

We also analyzed pipelines proposed in recent literature, including two-stage tumor filtering followed by One-vs-Rest classification with gradient-boosted aggregation [3]. While these approaches achieve strong results on large whole-slide datasets, they rely on millions of tiles and explicit tumor annotations.

Given the significantly smaller scale of our dataset and the absence of dense supervision, directly adopting such pipelines was impractical. Instead, we retained their core insight—separating coarse and fine-grained decisions—by implementing a hierarchical cascade of binary classifiers aligned with clinical taxonomy.

References

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